Analog Integrated Circuit Design

Cadence Tools

Lab 02

Common Source Amplifier

Contents

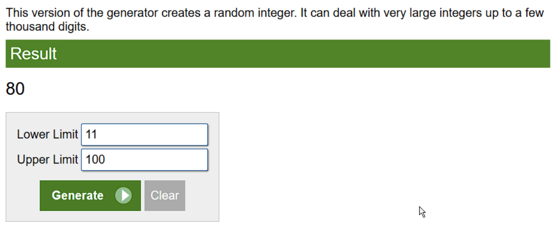
1 Part 1: Sizing Chart

1.1 Design Specifications

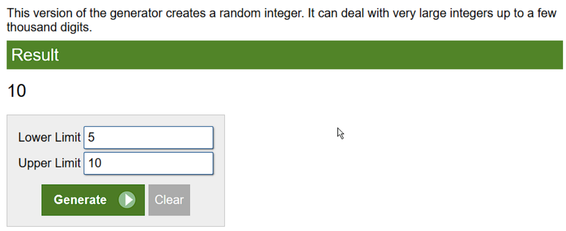
The target specifications for the 0.18 µm CMOS technology are:

Table 1: Design specifications.

Parameter	Value
DC Gain $\lvert A_v \rvert$	10
Supply V_{DD}	1.8 V
Current consumption I_{DQ}	80 µA
Channel length L	2 µm



(a) Random number generator showing bias current = 80 µA.



(b) Random number generator showing gain = 10.

Figure 1: Randomly assigned design parameters for this report.

1.2 Hand Analysis: Design Equations

1.2.1 Load Resistor R_D

To maximise output signal swing, the common-mode (CM) output is placed at $V_{DD}/2$, which means the DC voltage drop across R_D equals $V_{DD}/2$:

$$V_{R_D} = \frac{V_{DD}}{2} = \frac{1.8}{2} = 0.9\,\text{V} \tag{1}$$

Given the DC bias current $I_{DQ} = 80\,\mu\text{A}$:

$$\boxed{R_D = \frac{V_{R_D}}{I_{DQ}} = \frac{0.9}{80 \times 10^{-6}} = 11.25\,\text{k}\Omega} \tag{2}$$

1.2.2 Required V_Q^*

The small-signal voltage gain of the resistively loaded CS amplifier (ignoring r_o for now) is

$$|A_v| \approx g_m R_D = \frac{2 I_D}{V^*} R_D = \frac{2\,V_{R_D}}{V^*} \tag{3}$$

where $V^* = 2I_D/g_m$ is the effective overdrive (equal to V_{ov} only for a square-law device). Rearranging for the required V^*:

$$\boxed{V_Q^* = \frac{2\,V_{R_D}}{|A_v|} = \frac{2 \times 0.9}{10} = 0.18\,\text{V}} \tag{4}$$

1.2.3 Transistor Sizing: Width W

Because $I_D \propto W$ at any fixed V_{GS} (irrespective of the operating model), the sizing chart generated with $W_{\text{ref}} = 10\,\mu\text{m}$ gives a reference current I_{DX} at $V^* = V_Q^*$. The required width is then found by ratio and proportion:

$$\frac{W}{10\,\mu\text{m}} = \frac{I_{DQ}}{I_{DX}} \implies W = 10\,\mu\text{m} \times \frac{I_{DQ}}{I_{DX}} \tag{5}$$

1.3 Simulation Results: Sizing Chart

1.3.1 V^* and V_{ov} vs. V_{GS} where $V_{th} = 0.383V$

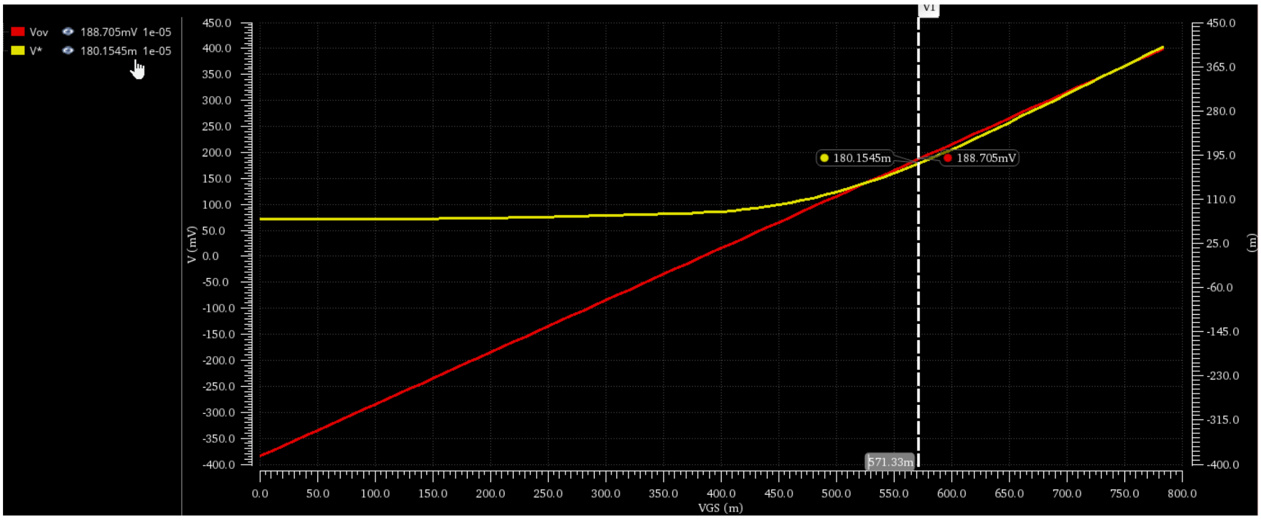


Figure 2: $V^* = 2I_D/g_m$ (Yellow) and $V_{ov} = V_{GS} - V_{TH}$ (Red) vs. V_{GS} for the NMOS with $W = 10\,\mu\text{m}$, $L = 2\,\mu\text{m}$.

1.3.2 I_D, g_m, g_{ds} vs. V_{GS}

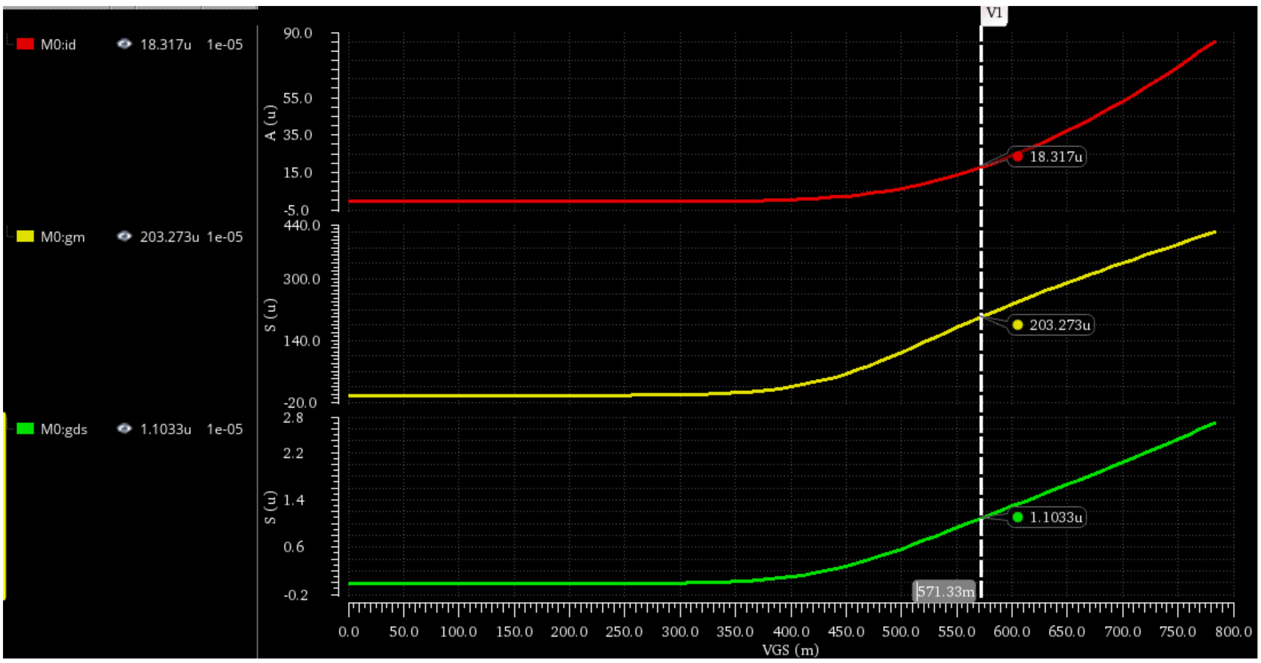


Figure 3: I_D , g_m , and g_{ds} vs. V_{GS} for the reference NMOS ($W = 10\,\mu\text{m}$, $L = 2\,\mu\text{m}$, $V_{DS} = V_{DD}/2$).

From the sizing chart (Figure 2), at $V_Q^* = 0.18\,\text{V}$ we read:

Table 2: Values read from the sizing chart at $V_Q^* = 0.18\,\text{V}$, $W_{\text{ref}} = 10\,\mu\text{m}$, $L = 2\,\mu\text{m}$.

Quantity	Symbol	Value
Reference drain current	I_{DX}	$18.32\,\mu\text{A}$
Reference transconductance	g_{mX}	$203.27\,\mu\text{S}$
Reference output conductance	$g_{ds,X}$	$1.103\,\mu\text{S}$
Vov_Q	$V_{ov,Q}$	$188.71\,\text{mV}$
Corresponding V_{GS}	V_{GSQ}	$571.33\,\text{mV}$

$$W = 10\,\mu\text{m} \times \frac{80\,\mu\text{A}}{I_{DX}} \approx 43.67\mu\text{m} \tag{6}$$

1.3.3 *Scaled Small-Signal Parameters*

Since $g_m \propto W$ and $g_{ds} \propto W$ at constant V_{ov}:

$$g_{mQ} = g_{mX} \times \frac{W}{10\,\mu\text{m}} \approx 887.68\mu\text{S} \tag{7}$$

$$g_{ds,Q} = g_{ds,X} \times \frac{W}{10\,\mu\text{m}} \approx 4.82\mu\text{S} \implies r_{oQ} = \frac{1}{g_{ds,Q}} \approx 207.6\text{ K}\Omega \tag{8}$$

1.3.4 *Gain Verification*

The actual gain including r_o:

$$A_v = -g_m(R_D \parallel r_o) = -887.68\mu \times \frac{11.25\text{k} \times 0.2076\text{M}}{11.25\text{k} + 0.2076\text{M}} = -9.5 \approx -10 \tag{9}$$

The parameters are correct!

Table 3: Final design parameters from Part 1 sizing chart ($0.18\,\mu\text{m}$ CMOS, $L = 2\,\mu\text{m}$, $I_{DQ} = 80\,\mu\text{A}$, $|A_v| = 10$).

Parameter	Symbol	Value
Transistor Width	W	$43.67\,\mu\text{m}$
Channel Length	L	$2\,\mu\text{m}$
Transconductance	g_m	$887.68\,\mu\text{S}$
Output Conductance	g_{ds}	$4.82\,\mu\text{S}$
Output Resistance	r_o	$207.6\,\text{k}\Omega$
Load Resistor	R_D	$11.25\,\text{k}\Omega$
Gate-Source Bias Voltage	V_{GSQ}	$571.33\,\text{mV}$

2 Part 2: CS Amplifier

2.1 OP and AC Analysis

2.1.1 DC Operating Point

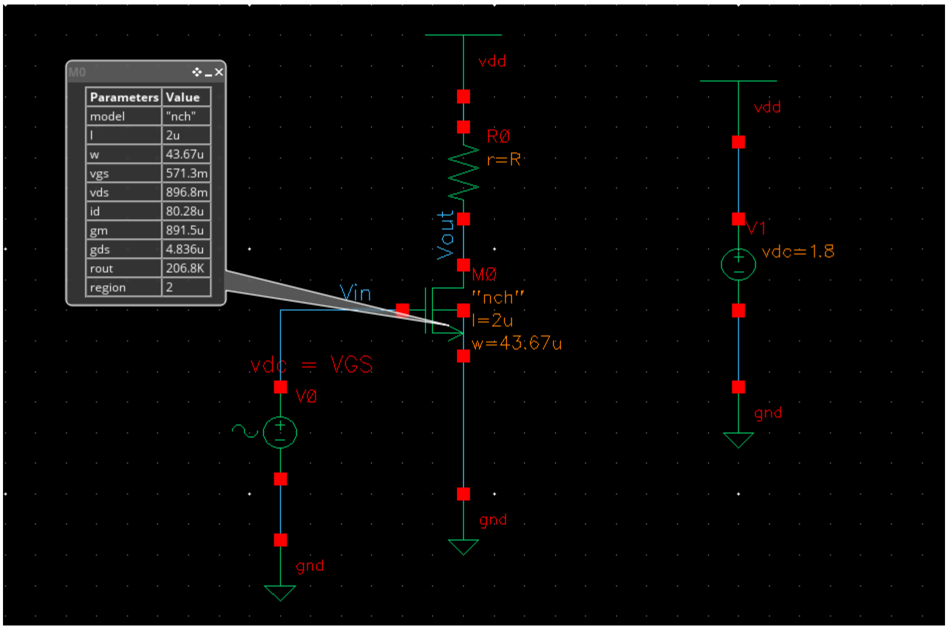


Figure 4: DC operating point annotation from Cadence Virtuoso.

Table 4: Comparison of key parameters between the sizing chart (Part 1) and DC operating point simulation (Part 2).

Parameter	Part 1 (Sizing Chart)	Part 2 (DC Simulation)	% Error
V_{GS}	571.33 mV	571.3 mV	–
I_D	80 μA	80.28 μA	0.35
g_m	887.68 μS	891.5 μS	0.43
g_{ds}	4.82 μS	4.836 μS	0.33
r_o	207.6 kΩ	206.8 kΩ	0.385
R_D	11.25 kΩ	11.25 kΩ	0.000
W	43.67 μm	43.67 μm	0.000
L	2 μm	2 μm	0.000

Comments

1) DC OP comparison with Part 1 (chart-based design):
Since the design used a chart extracted directly from simulation with the same technology and the same L, the operating point from the DC simulation matches the chart values closely. The drain current is approximately 80 µA, confirming that the width scaling by ratio and proportion was correct. The region code reads 2 (saturation), as required for a linear amplifier.
2) r_o vs. R_D – is ignoring r_o justified?would the error remain the same at minimum L?
From the DC OP, $r_{oQ} = 1/g_{ds,Q}$. At $L = 2\,\mu\text{m}$, r_o is typically much larger than $R_D = 11.25\,\text{k}\Omega$, so the error introduced by ignoring r_o in the gain formula $|A_v| \approx g_m R_D$ is small (a few percent). However, if minimum L were used (200 nm), r_o would drop significantly, and the parallel combination $R_D \| r_o$ would be noticeably smaller than R_D alone – the assumption would *not* be justified.

2.1.2 Intrinsic Gain and Amplifier Gain

The **intrinsic gain** of the transistor is defined as:

$$A_{\text{int}} = -g_m r_o = \frac{g_m}{g_{ds}} \approx -184 \tag{10}$$

This is the maximum possible voltage gain achievable from a single transistor (when $R_D \to \infty$, i.e., current-source load).

The **amplifier gain** with a resistive load is:

$$|A_v| = g_m(R_D \| r_o) = g_m \frac{R_D\, r_o}{R_D + r_o} \approx 9.5 \tag{11}$$

Since $r_o \gg R_D$ in the long-channel case, we have $R_D \| r_o \approx R_D$ and:

$$|A_v| \approx g_m R_D \ll g_m r_o = A_{\text{int}} \tag{12}$$

The amplifier gain is therefore **much less** ($\ll$) than the intrinsic gain in this design, because the resistive load R_D is much smaller than r_o.

2.1.3 AC Analysis Results

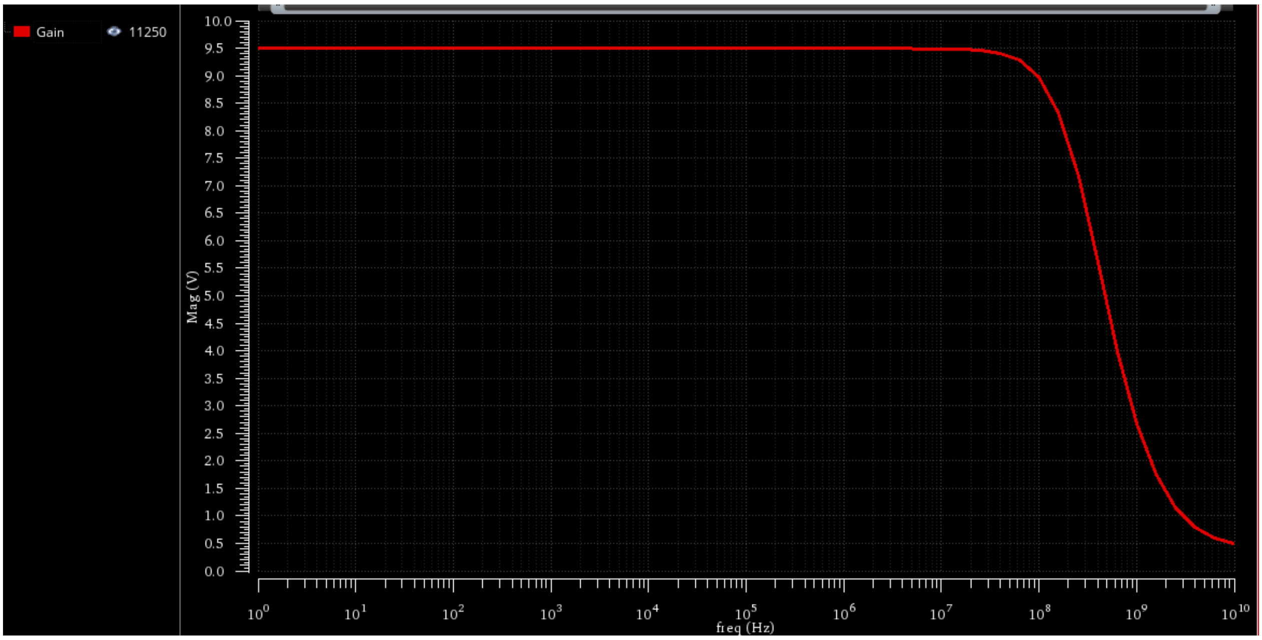


Figure 5: AC gain magnitude vs. frequency (1 Hz to 1 GHz).

ITI:LAB02_2:1	Gain	
ITI:LAB02_2:1	ymax(mag(VF("/Vout")))	9.512

Figure 6: Annotated DC gain value from Cadence.

Comments

The simulated DC gain from AC analysis is annotated as ≈ 9.578 (approximately 10), which agrees with the specification within reasonable tolerances of the chart-based design methodology. The gain rolls off at high frequency due to the parasitic capacitances at the output node (C_{gd}, C_{db}, and any load capacitance).

2.2 Gain Non-Linearity

2.2.1 DC Sweep: V_{OUT} vs. V_{IN}

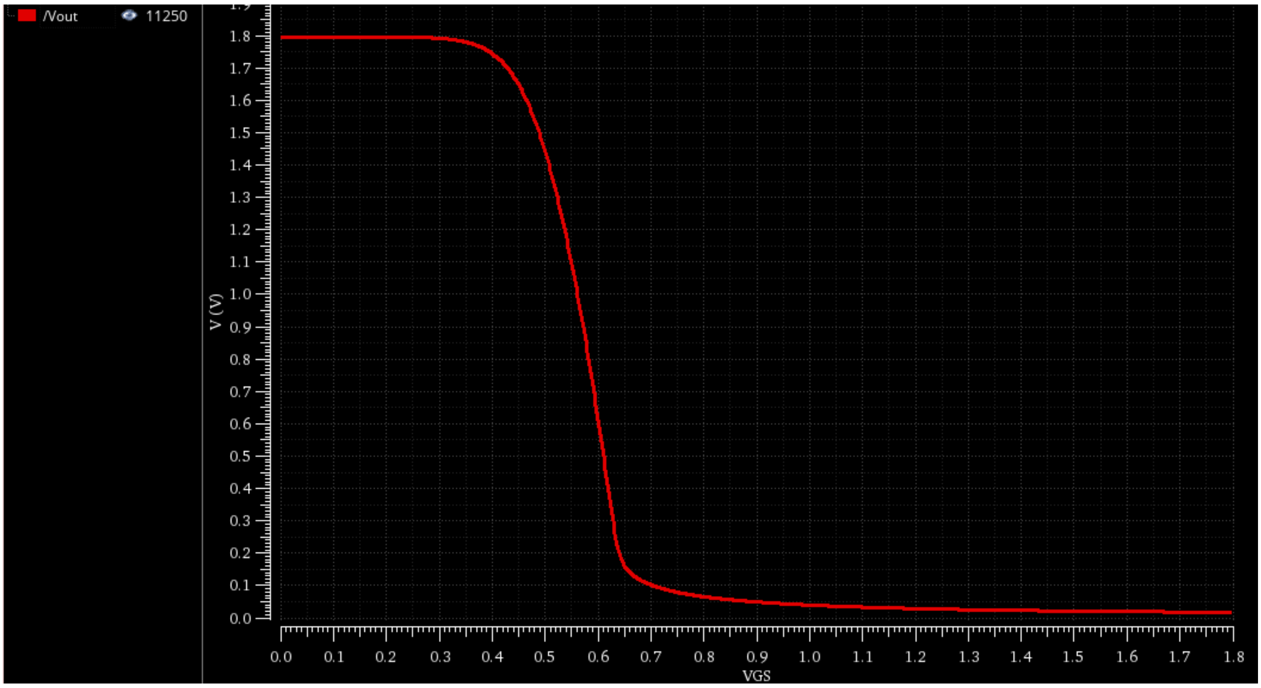


Figure 7: V_{OUT} vs. V_{IN} from DC sweep (0 to V_{DD}, 2 mV step).

Comments

Is the V_{OUT}–V_{IN} relation linear?
No. The transfer characteristic is not Linear. This is expected because:

- At low V_{IN} before V_{th} : the transistor is in cut-off; $V_{OUT} = V_{DD}$ (no current flows, no voltage drop across R_D).
- At the designed bias point $V_{IN} \geq V_{TH}$ and $V_{IN} \leq V_{OUT} + V_{TH}$: the transistor is in saturation; the curve has its maximum slope(linear), approximately equal to $-|A_v|$.
- At high V_{IN}, $V_{in} > V_{OUT}+V_{th}$: the transistor enters triode, V_{DS} drops, and the gain collapses(non linear). V_{OUT} approaches 0.

The relationship is only approximately linear (quasi-linear) over a small range of V_{IN} around the bias point V_{GSQ}.

2.2.2 Derivative: Small-Signal Gain vs. V_{IN}

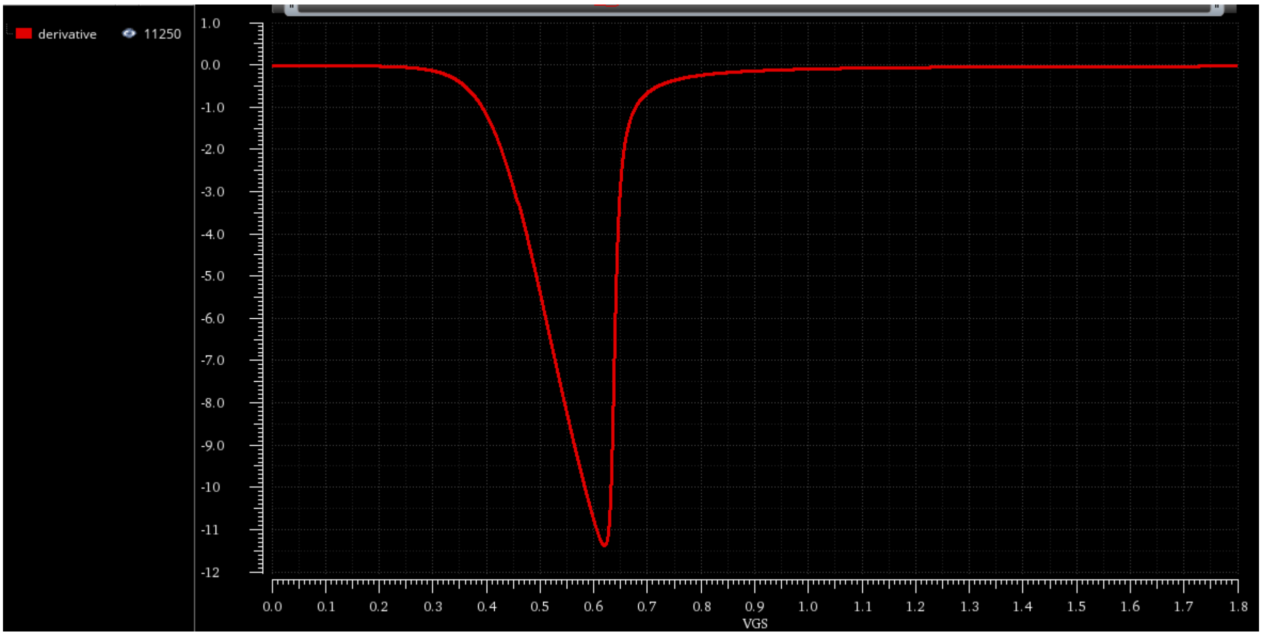


Figure 8: Derivative dV_{OUT}/dV_{IN} (instantaneous small-signal gain) vs. V_{IN}.

Comments

Is the gain linear (independent of input)?
No. The plot of dV_{OUT}/dV_{IN} vs. V_{IN} shows that the gain varies significantly with the input voltage. This is a direct consequence of the non-linear I_D–V_{GS} relationship of the MOSFET. Specifically:

- $g_m = \partial I_D/\partial V_{GS}$ is itself a function of V_{GS} (it is proportional to V_{ov} for the square-law device and saturates for short-channel devices).
- Since gain $= -g_m R_D$, any variation in g_m translates directly to gain variation.
- The gain is approximately constant only in a narrow range around the DC bias point.

2.2.3 Transient Simulation:

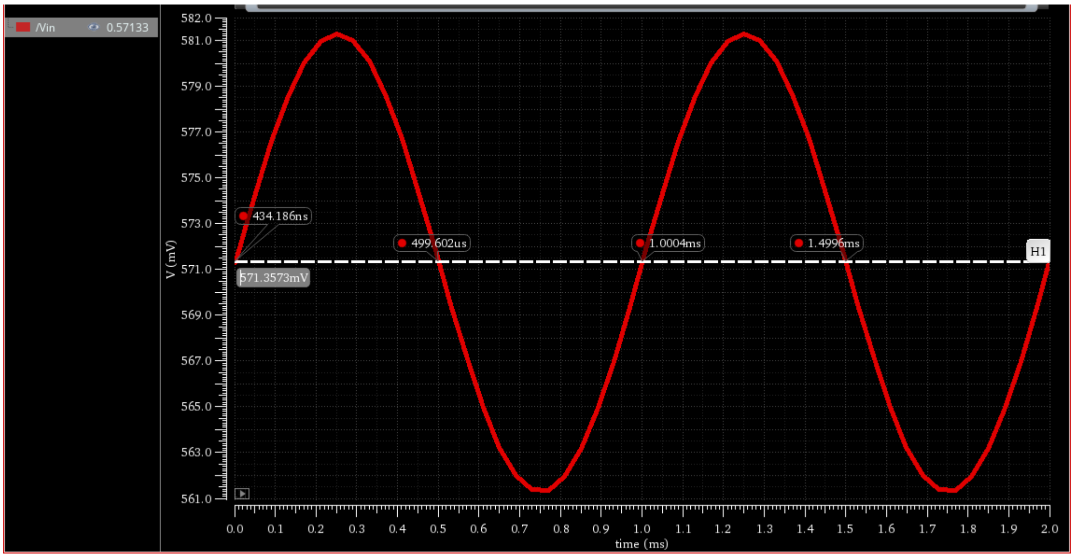


Figure 9: Input voltage V_{IN} vs. time.

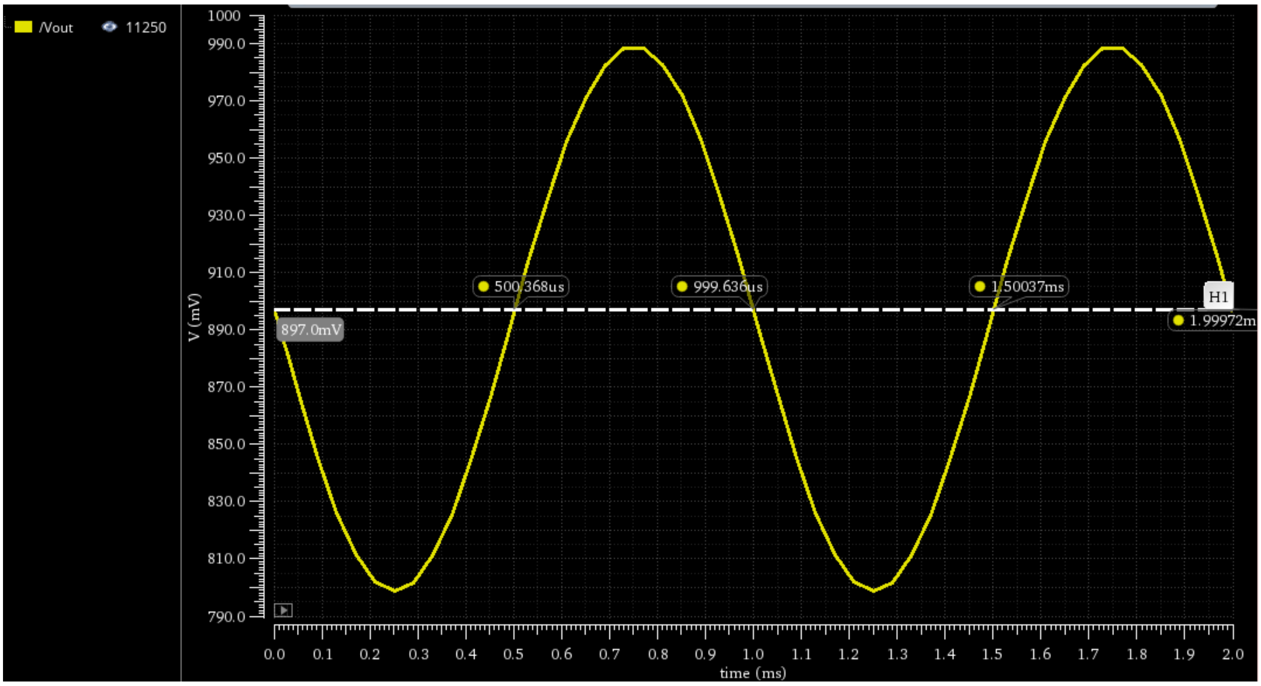


Figure 10: Output voltage V_{OUT} vs. time.

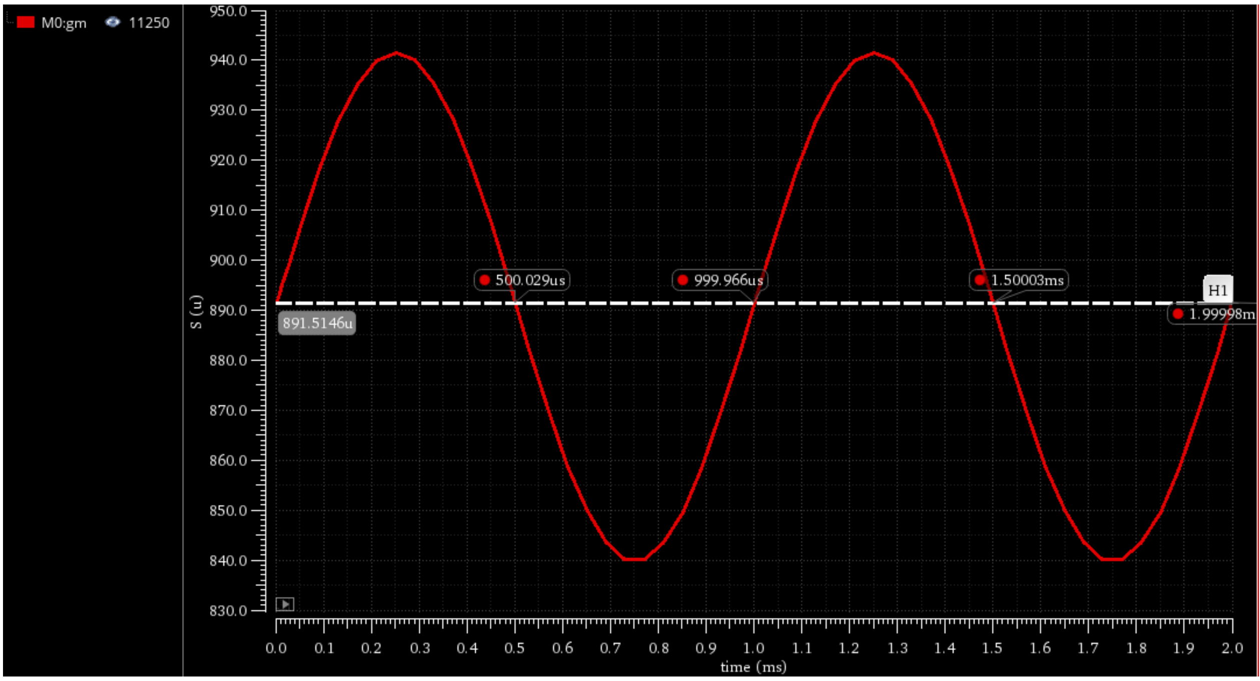


Figure 11: g_m vs. time under a 1 kHz, 10 mV sinusoidal input superimposed on V_{GSQ}.

Comments

Does g_m vary with the input signal?
Yes. The g_m vs. time plot shows that g_m oscillates sinusoidally around its DC bias value, in phase with the input sinusoid. This occurs because g_m depends on the instantaneous V_{GS}, which is the DC bias point plus the AC signal.
What does this mean? Is this amplifier linear?
A varying g_m means the instantaneous gain $A_v(t) = -g_m(t)R_D$ is time-varying. This introduces harmonic distortion: even a pure sinusoidal input produces an output that contains harmonics at $2f$, $3f$, etc. For a 10 mV amplitude the distortion is small (the amplifier is only mildly nonlinear), but it becomes severe for large signals. This amplifier is therefore **not linear** in the strict sense.

2.3 [Optional] Maximum Gain

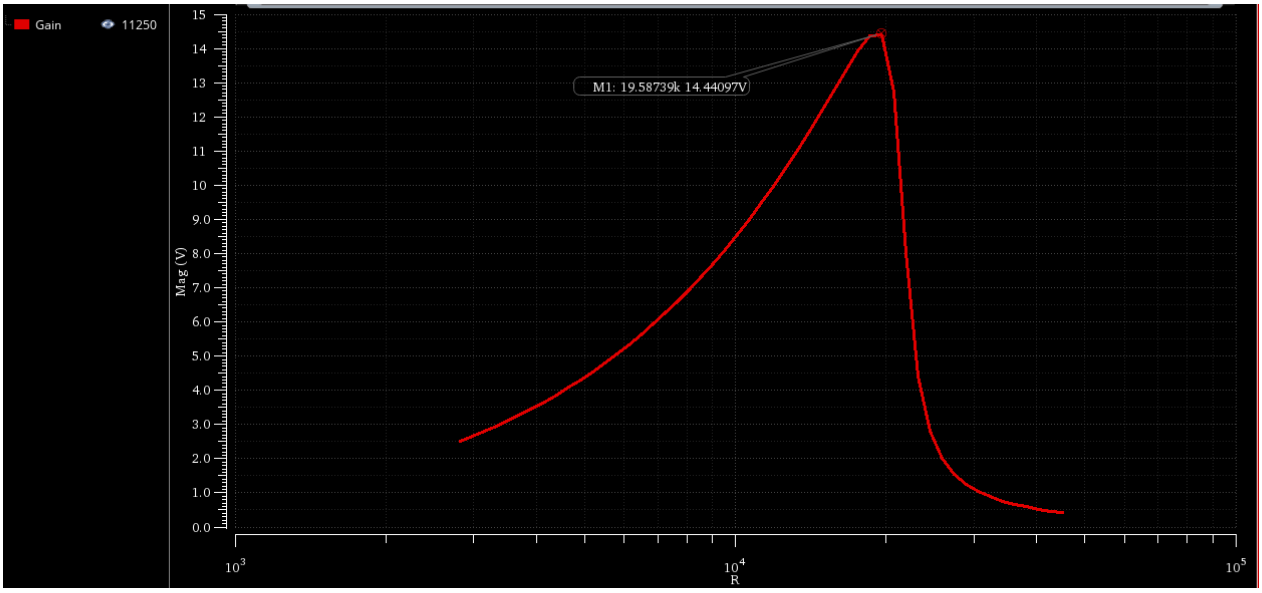


Figure 12: Gain vs RD

$$R_D^{(max)} = 19.587K\Omega \tag{13}$$

and the corresponding maximum gain is:

$$|A_v|_{max} = 14.44 \tag{14}$$

2.3.1 Hand Analysis: Gain as a Function of R_D

The small-signal gain including r_o is:

$$|A_v(R_D)| = g_m(R_D \| r_o) = g_m \frac{R_D\, r_o}{R_D + r_o} \tag{15}$$

Note, however, that **the bias current changes with** R_D when V_{GSQ} is held constant and the supply is fixed. With $V_{outCM} = V_{DD} - I_D R_D$, the bias point shifts as R_D varies; consequently g_m and r_o are themselves functions of R_D. This makes the analysis more involved. In the simplified case where g_m and r_o are treated as constants (fixed bias current):

- For $R_D \ll r_o$: $R_D \| r_o \approx R_D$, so gain $\approx g_m R_D$ increases linearly with R_D.
- For $R_D \gg r_o$: $R_D \| r_o \approx r_o$, so gain $\approx g_m r_o = A_{\text{int}}$, a constant.
- Maximum gain equals the intrinsic gain $A_{\text{int}} = g_m r_o$, achieved when $R_D \to \infty$.

In practice with a fixed V_{DD} and fixed V_{GSQ}, increasing R_D eventually pushes V_{DS} below V_{ov} (transistor leaves saturation), causing the gain to *drop*. So the gain peaks at an optimal R_D that keeps the transistor in saturation while maximising $g_m R_D$.

The condition for maximum gain under the constraint $V_{out,CM} = V_{DD} - I_D R_D \geq V_{ov}$ leads to the **maximum** R_D just before the transistor exits saturation:

$$R_D^{(max)} = \frac{V_{DD} - V_{ov}}{I_D} = 20K\Omega \tag{16}$$

and the corresponding maximum gain is:

$$|A_v|_{max} = g_m \, R_D^{(max)} \| r_o \approx g_m \, \frac{V_{DD} - V_{ov}}{I_D} = \frac{2(V_{DD} - V_{ov})}{V^*} = 18 \tag{17}$$

Table 5: Comparison of Simulation and Hand Analysis Results

Parameter	Simulation	Hand Analysis	Error
Maximum Load Resistance ($R_D^{(max)}$)	19.587 KΩ	20 KΩ	2.065%
Maximum Gain ($\|A_v\|_{max}$)	14.44	18	19.7%

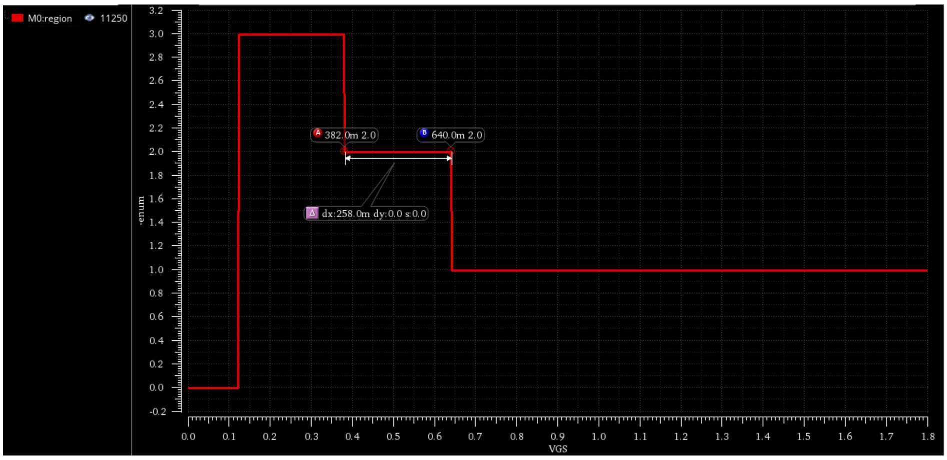


Figure 13: Region of NMOS vs VGS

the available swing eqauls **258 mV**.

Comments

Why does gain first increase then decrease with R_D?

- **Increasing phase (R_D small):** $R_D \ll r_o$, so increasing R_D directly increases $g_m R_D$. The transistor stays in saturation.
- **Decreasing phase (R_D large):** As R_D grows, $V_{DS} = V_{DD} - I_D R_D$ decreases. Eventually $V_{DS} < V_{ov}$, pushing the transistor into triode, where g_m drops sharply and g_{ds} rises. The gain collapses.

Effect of supply scaling on gain:
Scaling down V_{DD} reduces the headroom available for the output swing. Since the maximum gain is $\propto (V_{DD} - V_{ov})$, a lower supply directly reduces the achievable gain.

2.4 [Optional] Gain Linearization via Feedback

2.4.1 PMOS Active Load Schematic

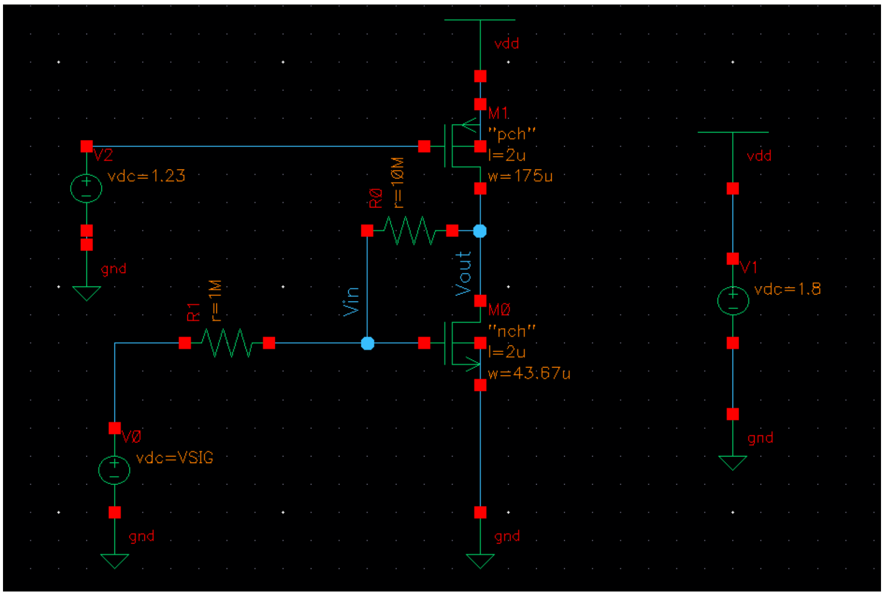
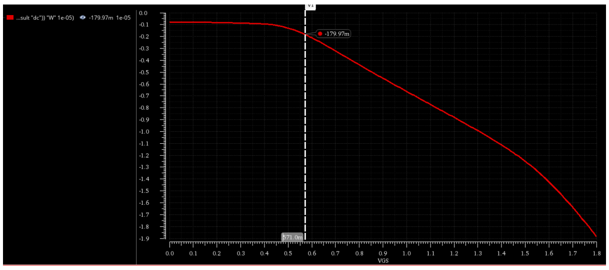
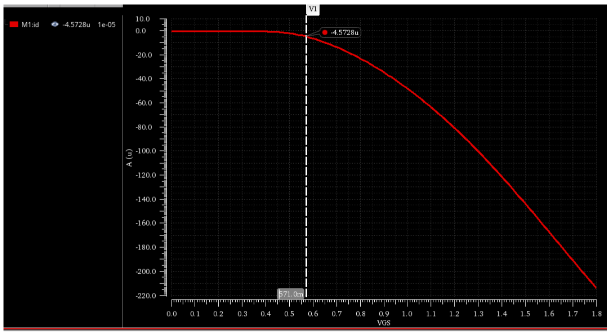


Figure 14: CS amplifier with PMOS current-source load and feedback resistors R_{in} and R_f for gain linearization.

2.4.2 PMOS Sizing Chart



(a) V^* vs. $|V_{GS}|$ for the reference PMOS ($W = 10\,\mu\text{m}$, $L = 2\,\mu\text{m}$).



(b) vs. $|V_{GS}|$ for the reference PMOS ($W = 10\,\mu\text{m}$, $L = 2\,\mu\text{m}$).

Figure 15: Characterization curves for the reference PMOS.

From the PMOS characterization charts, we extract the operating point values:

- $V_{GSX} = 571$ mV
- $I_{DX} = 4.5728\mu$A

2.4.3 PMOS Sizing

Using the PMOS sizing chart (Figure 15a), the same $V_Q^* = 0.18\,\text{V}$ as NMOS is assumed for the PMOS load. The PMOS must carry the same bias current $I_{DQ} = 80\,\mu\text{A}$:

$$W_P = 10\,\mu\text{m} \times \left.\frac{I_{DQ}}{I_{DX,P}}\right|_{V^*=0.18\,\text{V}} = 175\,\mu\text{m} \tag{18}$$

2.4.4 Feedback Resistors

The feedback resistor R_f sets the closed-loop gain:

$$|A_{v,\text{closed}}| \approx \frac{R_f}{R_{in}} \implies R_f = |A_v| \times R_{in} = 10 \times 1\,\text{M}\Omega = 10\,\text{M}\Omega \tag{19}$$

With $R_{in} = 1\,\text{M}\Omega$, choose $R_f = 10\,\text{M}\Omega$.

2.4.5 DC Sweep with Feedback

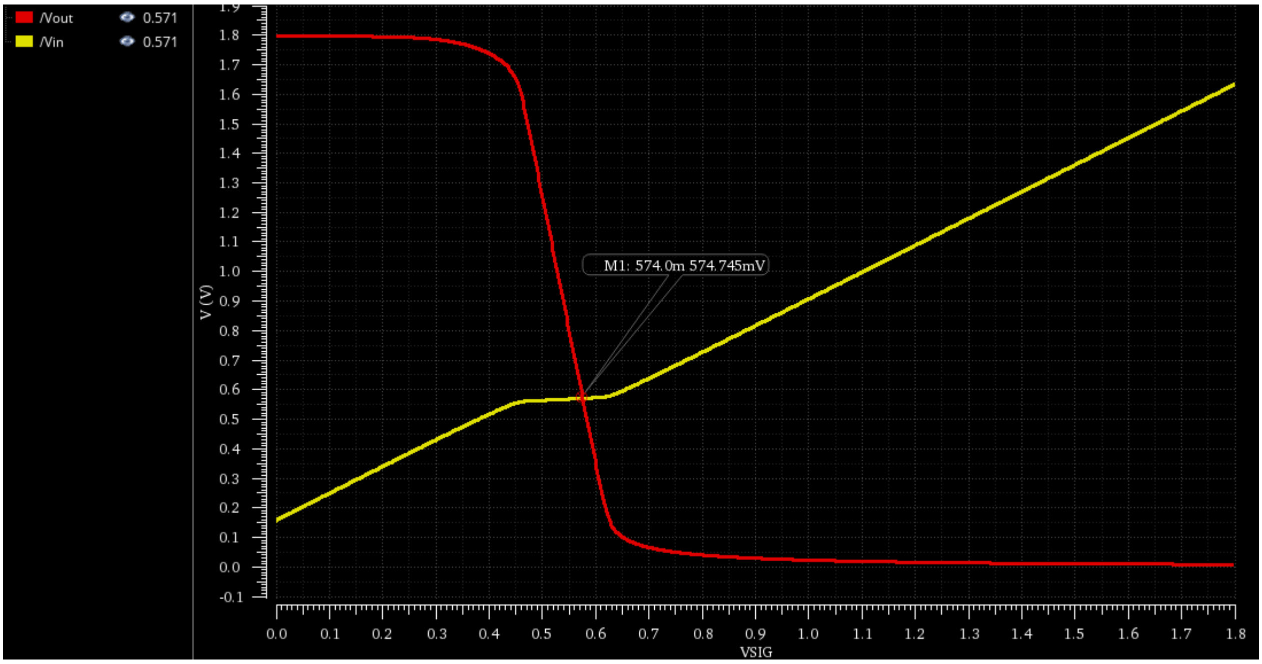


Figure 16: V_{IN} and V_{OUT} vs. V_{SIG} (DC sweep with PMOS load and feedback).

Comments

At what voltage do the V_{IN} and V_{OUT} curves cross?
They cross at $V_{SIG} = 574mV$ (the DC bias gate voltage of M1). $I_{DN} = I_{DP}$.
Is V_{OUT} vs. V_{SIG} more linear than before?
We can divide the relationship between V_{out} and V_{SIG} into three distinct operating regions:

- **Cut-off Mode:** When $V_{SIG} < V_{TH}$, the MOSFET is in cut-off. resulting in $V_{out} = V_{DD} \approx 1.8$ V.
- **Saturation Region:** When $V_{SIG} \geq V_{TH}$ and $V_{GD} \leq V_{TH}$, the MOSFET operates in the saturation region. Because feedback improves linearity, this can be approximated as a linear region (characterized by a high slope and high gain). As current flows through the transistor, V_{out} decreases due to the voltage drop across the active load.
- **Triode Region:** When $V_{SIG} > V_{TH}$ and $V_{GD} > V_{TH}$, the MOSFET enters the triode region. The gain drops significantly, resulting in a very small slope and minimal variation of V_{out} with respect to V_{SIG}.

Consequently, the relationship between V_{out} and V_{SIG} is not strictly linear across the entire curve.

2.4.6 Derivative (Gain) with Feedback

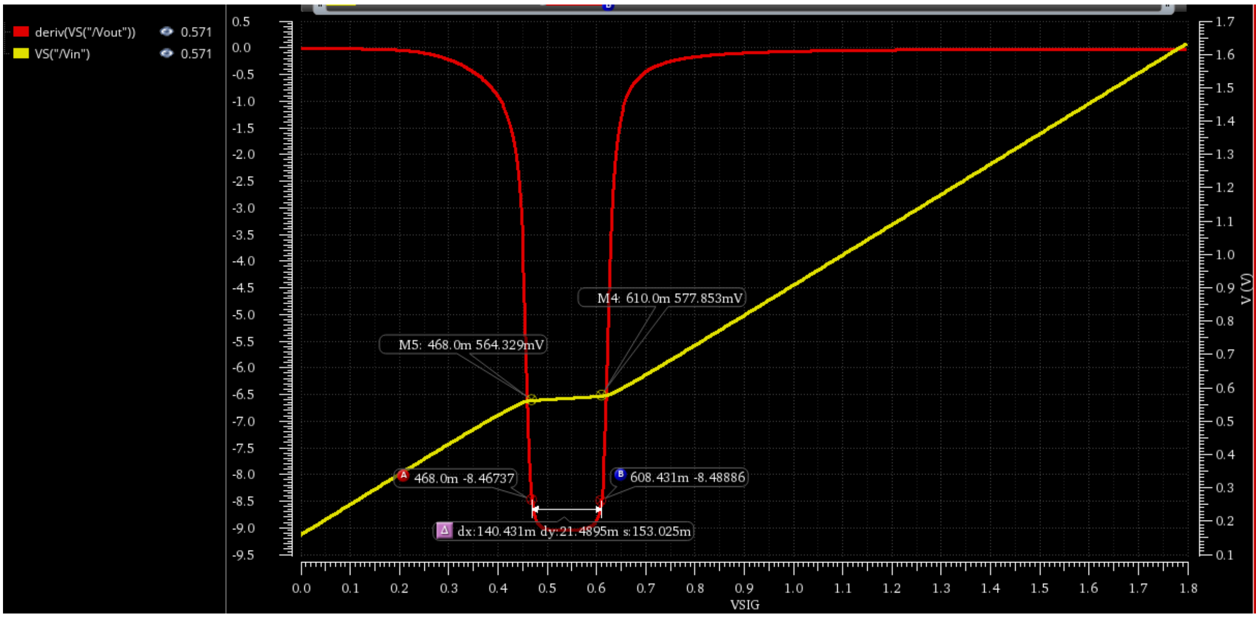


Figure 17: dV_{OUT}/dV_{SIG} (small-signal gain) vs. V_{SIG} with feedback.

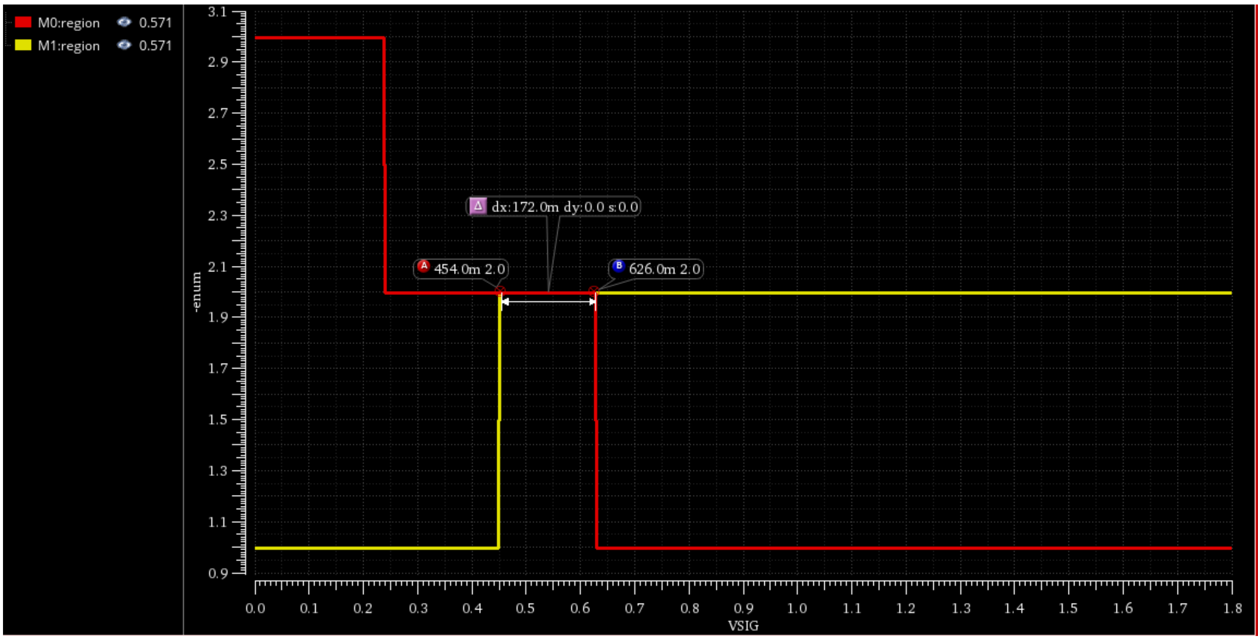


Figure 18: Regions of operations for NMOS and PMOS

Comments

Is the gain more constant with feedback?
Yes. The derivative plot shows a much flatter gain characteristic compared to the open-loop case. Feedback reduces gain non-linearity at the cost of reduced gain magnitude. The linear input range is extended because the feedback desensitizes the closed-loop gain to variations in the open-loop gain A_{OL}: as long as $A_{OL} \gg R_f/R_{in}$, the closed-loop gain $\approx R_f/R_{in}$ regardless of g_m fluctuations.
Analytical linear input range:
The amplifier fails when M1 or M2 exits saturation ($V_{DS} < V^*$). The output range before clipping is approximately $V_{DD} - 2V^* = 1.8 - 2(0.18) = 1.44\,\text{V}$. The corresponding input range is:

$$\Delta V_{in} \approx \frac{V_{DD} - 2V^*}{|A_v|} = \frac{1.44}{10} = 144\,\text{mV} \tag{20}$$

The range for V_{SIG} where linear gain is from 468 mV to 610 mV is 0.142 V.
The range for V_{IN} where the gain is linear is from 564 mV to 577 mV is 13 mV and average value for V_{in} is 618 mV (it's the region where the derivative of V_{out} is constant). For NMOS to be in sat:

$$V_{GS} \geq 382.6 \text{ mV} \quad \text{and} \quad V_{out} \geq V_{ov} = V_Q^* = 180 \text{ mV}$$

$$\mathbf{V_{out}} \geq \mathbf{0.18} \text{ V}$$

For PMOS to be in sat:

$$\begin{aligned} V_{SD} &\geq V_{SG} - |V_{THP}| \\ V_{out} &\leq V_G + |V_{THP}| \\ V_{out} &\leq 1.2 + 0.393 \end{aligned}$$

$$\mathbf{V_{out}} \leq \mathbf{1.593} \text{ V}$$

So the range of V_{out} for the linear gain is 1.413 V = 1.593 - 0.18 , then the range of the input that will be given by

$$\text{Range } V_{SIG} = \frac{\text{Range } V_{OUT}}{\text{Gain}} = \frac{1.413}{10} = 0.1413 \text{ V}$$

Using rule,

$$\text{Range } V_{SIG} = \frac{V_{DD} - 2V_Q^*}{\text{Gain}} = \frac{1.8 - 2 \times 0.18}{10} = 0.144 \text{ V}$$

$$\text{Error\%} = \frac{0.144 - 0.142}{0.144} \times 100 = 1.39\%$$